Guidelines for Evidence Collection, Preservation and Transportation

Audio-Visual Analysis

- All items should be packaged in containers of suitable size.
- All items should be packaged in containers that will prevent contamination or deleterious change.
- Ensure that all evidence collected is properly documented, labeled, marked, photographed and inventoried before it is packaged.
- Remember that evidence may also contain latent, trace, or biological evidence so take appropriate steps to preserve it.
- Package all digital evidence in anti-static packaging to prevent it from static electricity. Only paper bags and envelopes, cardboard boxes and antistatic containers should be used for packaging of digital evidence.
- The video evidence should be collected in its original format as it is recorded on the recording device (DVR, VCR, etc.).
- Make sure that the Pin/Pattern/password of the digital device (DVR, Mobile Phone etc.) is being provided by the evidence submitting person / Agency.
- Evidence should be packaged in a manner to avoid getting bent, scratched or otherwise deformed. Plastic material should not be used for packaging.
- Collect all power supplies, cables and adapters for all electronic devices seized.
- Shock resistance packaging material should be used to avoid physical damage to any components of the device(s).
- Label all containers used to package digital evidence clearly and properly.
- The packaging areas should be void of ultraviolet (UV) light (present in some types of fluorescent tubes). UV may hasten the degradation process.
- The packaging environment should have a mild temperature and humidity. An extreme environment can lead to spoliation of potential evidence, for example mold growth.
- All items are packaged in containers that can be sealed.
- The seals must display the initials of the personnel creating the seal.
- The seals must display the date when the seal is created.

- The seals must be made from a material that is tamper evident. The removal of the seal must cause some visible damage to the container that can indicate that the seal has been removed or tampered with.
- The evidence packaging is labeled with at least the Submitting Agency case number and item number, date and initials of the person who packaged.
- It is advisable that forensic evidence tape shall be used.

Transportation

When transporting audio/video evidence:

- Keep the evidence away from magnetic fields such as those produced by radio transmitters, speaker magnets, and magnetic mount emergency lights. Other potential hazards are seat heaters and any device or material that can produce static electricity.
- Avoid keeping the evidence in a vehicle for prolonged periods of time. Heat, cold, and humidity can damage or destroy the evidence.
- Ensure that computers or electronic devices are packaged and secured during transportation to prevent damage from shock and vibration.
- Document the transportation of the evidence and maintain the Chain of Custody on all evidence transported.

Storage

When storing audio/video evidence:

- Ensure that the digital evidence is inventoried.
- Ensure that the digital evidence is stored in a secure, climate-controlled environment or a location that is not subject to extreme temperature or humidity.
- Ensure that the evidence is not exposed to magnetic fields, moisture, dust, vibration, or any other elements that may damage or destroy it.

Computer Forensic

- All items should be packaged in suitable sized containers that will prevent contamination or deleterious change.
- Ensure that all digital evidence collected is properly documented, labeled, marked, photographed and inventoried before it is packaged.
- Remember that digital evidence may also contain latent, trace, or biological evidence and take the appropriate steps to preserve it.

- Package all digital evidence in anti-static packing to prevent it from static electricity. Only paper bags and envelopes, cardboard boxes and antistatic containers should be used for packaging of digital evidence.
- Evidence should be packed in a manner to avoid from being bent, scratched or otherwise deformed. Plastic material should not be used for packing.
- Collect all power supplies, cables and adapters for all electronic devices seized.
- Shock resistance packing should be used to avoid physical damage to any component of the device(s).
- Label all containers used to pack digital evidence clearly and properly.
- Main system units and/or notebooks need to be secured in an appropriate container to avoid tampering or spoliation of the potential digital evidence that could reside in it.
- The packing areas should be void of ultraviolet (UV) light (present in some types of fluorescent tubes). UV may hasten the degradation process.
- The packing environment should have a mold temperature and humidity. An extreme environment can lead to spoliation of potential evidence, example mold growth.
- The collected digital device(s) should be stored in a secure environment or location that is not subject to extreme temperature or humidity. It should not be exposed to magnetic fields, dust, vibration, moisture or any other environmental elements that may damage it.
- Leave Mobile Devices/ Smart Phones in the power state (On or off) in which they are found. If possible, place the phone in flight or airplane mode.
- Mobile Devices/ smart phones should be isolated from the Network using Network Isolation Techniques i.e. Faraday Isolation bags, Radio Frequency shielding material, anti-static packing and aluminum foils.
- All items are packed in containers that can be sealed.
- The seals must display the initials of the Submitting Agency Personnel, creating the seal.
- The seals must display the date when the seal is created.
- The seals must be made from a material that is tamper evident. The removal of the seal must cause some visible damage to the container that can indicate that the seal has been removed or tampered with.
- The evidence packing is labeled with at least the Submitting Agency case number and item number.

Transportation

When transporting digital evidence:

- The Potential Digital Evidence should not be left unattended during the transportation process.
- The Digital Evidence First Responder should maintain the Chain of Custody throughout the transporting process to prevent possible tampering or spoliation, and maintain the integrity and authenticity of the digital devices and evidence.
- Keep digital evidence away from magnetic fields such as those produced by radio transmitters, speaker magnets, and magnetic mount emergency lights. Other potential hazards are seat heaters and any device or material that can produce static electricity.
- Avoid keeping digital evidence in a vehicle for prolonged periods of time. Heat, cold, and humidity can damage or destroy digital evidence.
- Ensure that computers and electronic devices are packed and secured during transportation to prevent damage from shock and vibration.
- Document the transportation of the digital evidence and maintain the Chain of Custody on all evidence transported.

Firearms and Tool Marks

In order to minimize safety risks and contamination of evidence the following measures should be followed while packing the evidence:

- Every evidence exhibit must be packaged separately.
- Every firearm must be packaged in unloaded condition with safety on.
- There must not be live rounds in the chamber of the firearm, magazine or in the parcel.
- Every cartridge case and bullet must be packaged separately.
- Evidence submitted for Gun Shot Residue (GSR) analysis must be packaged in hard box instead of cloth bag or paper envelope. Layers of the clothes containing GSR must not touch with the other layers. Clothes must be wrapped by placing a white paper sheet between the layers of clothes before packing it in a hard box.
- For serial number restoration of firearms, area containing obliteration should be marked clearly if there is more than one location of obliteration.
- For trajectory analysis, vehicles must not be washed or cleaned at all prior to examination. Suspected bullet holes must be covered with white paper.
- Seals must be intact and as per mentioned in the docket.

- If firearm is recovered from water or any other liquid, then submit the firearm with the same sample of water or liquid from which it has been recovered.

If any evidence related to a particular case is previously in PFSA custody, clearly mention the link of previous evidence while submitting new evidence.

Latent Finger Prints

Always use Personal Protective Equipment (PPE) while dealing with latent fingerprints evidence. It includes disposable gloves, facemask etc. In many instances, latent fingerprints can and should be developed at the crime scene by evidence technicians or crime scene search officers using a multitude of processes on all type of surfaces. Latent prints developed through traditional powder processing methods should be first photographed and then lifted with transparent tape and submitted to the laboratory. Detailed information concerning the case, date, location and orientation of the latent should be recorded on the lift card.

If latent prints at a crime scene appear to be visible (patent prints), or if the lifting process may pose unique challenges, the latent prints should be photographed. However, if any item of evidence is to be submitted to the lab for processing, it is best not to attempt any field recovery of latent prints.

Item – Non-porous or Non-absorbent surfaces (Glass, Metal, Tile, etc. may be processed in the field.)

Method –Generally, fingerprint powders should be used. Black powder is preferred because it produces the best ridge detail and is easier to compare. For powders to be used, the surface must be dry. Wet items may be processed with small particle reagent (SPR) or should be allowed to fully air-dry. The use of a hair dryer may produce too much heat causing the moisture in the latent print to evaporate.

***Reminder:** Whenever possible, non-porous items should be processed at the crime scene and the processed latent print(s) photographed / lifted and submitted to PFSA for further enhancement and comparison.

Discussion – Unnecessary transportation and handling may damage or even destroy print(s). In some cases, Cyanoacrylate Ester (commonly referred to as Super Glue Fuming) may be considered. This technique has proven successful in developing latent prints on items such as plastic baggies, Firearms, Styrofoam, and some types of leather.

Packaging of Non-Porous Items

Non-porous items should be packaged in such a way that they remain in fixed position and should not move freely inside the package during handling and transportation. Use of cardboard boxes and plastic ties is recommended as packaging material.

Item – Porous or absorbent surfaces (Paper, Untreated Wood, Cardboard, etc.)

Method – Generally, a variety of chemical processes are available. The photography of chemically developed latent prints is essential. Prints may fade or even completely disappear from the surface.

Examples of Chemical Processes

- Indanedione, Ninhydrin, Physical Developer for wet porous surfaces, Amido Black for bloody fingerprints etc.

Packaging of Dry Paper Items

- Dry paper items can be collected and sealed into plastic bags (zip-lock).

Packaging of Wet Paper Items

- Wet paper items should be air dried and once dried can be packaged.

Important Considerations For Print Collection

For visible prints on small objects, such as a cup, collect the entire object. If the item is fixed and not transportable then it must be processed for development of latent fingerprints at crime scene. Any bloody fingerprints on door etc. should be photographed with scale, before and after applying any blood enhancement technique e.g. Amido black etc.

Discussion – Photographs are important because damage to the impression may occur during attempts to enhance and lift it.
Avoid pressing or touching the impression with your finger or any object to see if the substance is dry or tacky. Doing so may result in damage to the print.

Case Submission

- Indicate all requested forensic examinations on the Evidence Submission Form. If it is a re-submission, note the previous PFSA case number in the appropriate space on the Evidence Submission Form.
- Do not process any item that you are planning to submit to the laboratory and do not place tape over items of evidence where you think there might be latent prints.
- Ensure that sharp objects such as broken glass or knives are packed safely and are properly labeled.

NOTE: Paper bags are not considered to be good packing materials for sharp or broken objects. Sharp objects can easily puncture the bag and cause injury.

- Good quality known prints are important and necessary. Smudged or blurred prints, overlays, too much ink, prints outside the blocks or off-centered, etc., will reduce the chances for an identification to be affected.

- If suspects are known, please obtain a set of fingerprint and palm print cards and submit them with the evidence.
- Take elimination fingerprints of the victims, family members, caretakers, etc.
- Original questioned document must be submitted for Fingerprint Examination.
- Docket / Cover letter addressed to the Director General, PFSA, Lahore clearly indicating required analysis and details of questioned and reference thumb impressions shall be required.
- Case Fee is required in Civil Cases hence, Bank Draft/ Pay Order from any bank in favor of DG, PFSA, Lahore must be submitted along with case (photocopy of bank draft is not acceptable).
- No case fee is required for Criminal Cases from Punjab province but a copy of FIR must be submitted.
- No case fee will be charged from departments under Punjab Govt.

Forensic Pathology

Mini Autopsy means examination of all viscera (heart, lungs, liver, spleen, kidneys, gastrointestinal tract and brain etc.), along with tissues of special interest (e.g. neck tissues in cases of strangulation, tissues around the bullet tract with tissues from exit and entry wound in gunshot cases, etc.). Mini autopsies constitute a QC and QA procedure for autopsies conducted in 800 autopsy centers of Punjab and elsewhere, where Forensic Histopathology services are not available. Approx. 1/3rd of all Medico legal autopsies are referred for Histopathological examination of the tissues, to reach the final diagnosis as to cause and manner of death, injuries inflicted during life or after death; and to ascertain the role of various contributory factors in the process of death. More than 2500 such cases are referred per year. Major problems encountered in such tissues sent from outside, are briefly mentioned here:

Poor Fixation

- The tissues are fixed in formalin solution, which is formaldehyde gas dissolved in water. With time, its concentration declines gradually, especially if the lid of container is not tightly closed.
- If the tissues are sent in formalin which is below 10%, the tissues get autolysed. Therefore, good quality, freshly prepared formalin should be used to fix the tissues.
- Fixative has to be added even in Exhumation cases
- Every specimen including soft tissues, bones, teeth and fetus etc. should be fixed

- If cytological examination of fluids, secretions and blood is required, then add few drops of 10% Formalin in the specimen
- Brief medical history of the deceased should be clearly mentioned in the forwarding letter.

Packing of Histopathology (Tissue) Samples

- Completely immerse the tissues into 10% formalin solution in a plastic jar having screwed lid.
- Quantity of formalin solution should be 3 – 4 times the tissue size.
- Tightly close the lid.
- Place evidence tape around the lid.
- Sign the evidence tape at regular intervals so that half part of the signature is on the evidence tape and the other half of the signature is on the container.
- Place stamps on the evidence tape in a similar manner. If evidence tape is not available, stamped red wax seals may be used as an alternative.
- Place the sealed container/jar in a plastic bag and tie the knot.

Labeling of Samples

Mention following information on the label on sample jars:

- Name of the deceased
- PMR/Case number
- Sample details
- Date and Time of sampling
- Collectors name, designation and signature

Transportation of Samples

- Place the sealed jar/s in an appropriately sized card board box and secure the containers in the box.
- Mention upper side on the box.
- Apply evidence tape at all opening slots of the card board box.
- Sign and stamp the evidence tape as mentioned above.
- Attach chain of custody form with the box.

- The sample jars may be transported individually as well. However, make sure that during transportation, they are kept in upright position, so that formalin is not drained out. Otherwise, tissues would get dry and autolysed.
- Put postmortem report, all relevant documents such as Report of Death (Report e Marg), FIR/ Application ”Rapat”, MLC, Road Certificate, Concise Case Details (Mukhtasar Halat e Muqadma), and in cases of Exhumation Legible copy of Magistrate or Court Order Relative/Family request along with sample of evidence tape and/or sample of signatures and stamp in an envelope.
- Seal the envelope with evidence tape signature and stamp as described above. If evidence tape is not available, stamped red wax seal may be used alternatively, as described above.
- Send the histopathology samples and documents to PFSA.

Polygraph Examination

Following documents are required for polygraph examination

- Copy of CNIC of suspect or attested photograph of suspect
- Copy of FIR
- Request letter for polygraph examination
- Suspect must have had proper breakfast and sufficient sleep
- Suspect should not have been tortured for last 24 hours
- Suspect should not have used any illegal drugs for at least 24 hours
- Suspect should not have any injury or physical illness
- Suspect hands must be washed
- Do not tell the suspect about polygraph examination
- Investigation officer is supposed to prepare the case fully before briefing

Questioned Documents

- Package the questioned document evidence in paper envelope of appropriate size and do not fold the questioned documents.
- Write the necessary information on the envelope before packaging the questioned document evidence in it. Do not write anything on the envelope after the evidence has been packaged.

- If the questioned document evidence is requested for indented writing/latent fingerprints test, then package the evidence carefully in such a way that it is not rubbed with other packaged documents.
- Envelopes used for packaging the evidence should protect the evidence from wear and tear and contamination.
- Make sure that all the necessary documents required for the case are attached with and properly documented.
- Case documents should be protected from severe environmental conditions such as moisture and fire.
- If the questioned evidence consists of charred or water-soaked documents, then pack them in a suitable hardboard box/container packed with cotton cushion so as to protect them from further destruction.
- Dispatch /submit the evidence in properly sealed form.
- Docket / cover letter should be addressed to the Director General, Punjab Forensic Science Agency, Lahore with clear mention of required examination. The questioned, routine/ admitted and dictated exemplars should be clearly marked and mentioned in the docket/cover letter.
- The case fee (If applicable) should be submitted only through **Bank Draft/Pay Order** in favor of the **Director General, Punjab Forensic Science Agency, Lahore**.
- In case of extra ordinary large number of questioned exhibits, only probative evidence items, as determined in consultation with the investigation officer, shall be examined.

Trace Chemistry

Due to the wide variety of evidence brought to the Trace Chemistry Department there is no single way to collect and pack the evidence. Each scene should be carefully examined for the presence and identification of trace evidence. The probative evidence should be collected in such a manner that it is not contaminated or lost. No harm to the integrity of the evidence should take place.

Evidence Collection Methods

Trace evidences like hair, fiber, paint chips, adhesive tapes etc. can be collected by the following methods.

- In all cases, the container must be labeled with at least the case number, exhibit number, date and initials of the concerned person, item number, and item description.

Handpicking

- Use forceps or other suitable tool to gently grasp the evidence item and carefully remove the item from the substrate.
- Package and seal the evidence item in a suitable container so that no contamination or deleterious change can occur.

Tape lifting

- Remove the first several inches of clean transparent adhesive tape from the roll to eliminate any possible environmental contamination.
- Obtain a section of tape from the roll. The size of the section of tape needed depends on the size of the item being examined.
- One or both ends of the tape are folded upon itself to establish handles from which the tape can be pulled away from the storage backing.
- The section of tape is applied to the item being examined. The adhesive side of the tape will collect any loosely adhering trace evidence.
- The collected tape lift is applied to a storage backing (e.g. clear acetate sheet).
- Package and seal the tape lift in a suitable container so that no contamination or deleterious change can occur.

Shaking

- A section of examination paper is placed under the item to be examined.
- The evidence item is shaken over the section of examination paper.
- The section of examination paper is visually examined for the presence of evidentiary material (e.g. hair, fibers, and paint chips). Any evidentiary material observed on the examination paper is removed by handpicking.
- The debris on the section of examination paper is transferred to a container and sealed so that no contamination or deleterious change can occur.

Scraping

- A section of examination paper is placed under the item to be examined.
- The evidence item is scraped with a clean scraping tool over the section of examination paper.
- The section of examination paper is visually examined for the presence of evidentiary material (e.g. hair, fibers, and paint chips). Any evidentiary material observed on the examination paper is removed by handpicking.

- The debris on the section of examination paper is transferred to a container and sealed so that no contamination or deleterious change can occur.

Paint Evidence

Paint samples collected should represent all the layers of the paint present. The sample should be chipped off down to the unpainted surface.

- If possible, submit the entire object on which the paint is observed, including smears and transfers. DO NOT attempt to remove paint from clothing, tools or objects where smears and transfers are deposited.
- If it is not feasible to submit the entire object, use a clean knife blade or scalpel to remove the area of interest including all the layers possible.
- Small samples can be retrieved using forceps or tweezers.
- Place sample in a paper fold or vial. DO NOT use an envelope. Small samples may be lost among the folds, openings and seals of the envelope.
- Place different samples in separate containers to avoid contamination.
- Be sure to seal the container and record the proper identifying information on the container and exterior packing.

Reference Comparison Sample for Paint Analysis

Collect a paint standard. A paint standard is a known sample of the undamaged paint collected from the same area as that of the damaged paint being analyzed.

- Standard paints should be at least ½ square inch of solid paint with all layers represented (down to the substrate).
- Take standard paint samples from near the damaged areas. Paint may vary in type or composition in different locations on a vehicle or item even though the color appears to be the same. Therefore, it is important that known paint standards be collected from each separate panel or area of the object showing fresh damage.
- Place each paint standard in a different paper fold, seal and label.
- In addition to the case and investigator information, the label must include the specific source of the sample e.g., make and model of the vehicle, known Paint samples must be collected from every vehicle or painted object involved in the incident, even if some known paint standard is included during the removal of questioned transfers.

Hair Evidence

For the majority of cases, the Trace Chemistry Department will evaluate human hair evidence to determine the potential for obtaining a DNA profile. Such an evaluation includes examining the hair characteristics to determine animal versus human; body origin (scalp, pubic, etc.) and growth phase.

Hair evidence can be collected in a number of ways including the following methods:

- Picking (For visible hair)
- Tape lifts
- Scraping
- Shaking

If the entire object, such as an article of clothing, containing possible hair evidence is to be submitted to the lab, place the object onto clean craft paper and paper fold. Seal the fold and place in a paper bag or envelope. Seal the container and include the proper identifying information.

Hair Sample Standards

Whenever hair is collected, the roots should be included because considerable information can be obtained from the root material.

Head or Scalp Hair: The hair should be representative of the center, front, back (including nape of the neck), and both sides of the scalp. Approximately 50 head hair should be collected. The sample should include both pulled and combed hair and include any variations in color and length. If additional facial hair are collected (i.e. sideburn or beard hair), these should be packed separately.

Pubic Hair: When indicated by the circumstances, collect pubic hair. Approximately20-30 pubic hair should be collected.

Animal Hair: Comb and pull hair; pulling is necessary as roots are needed for species identification in some animals. While a minimum number of hair is difficult to determine, good judgment should be used in collecting enough hair to represent the various types and colors of hair found on the animal. Hair should be collected from various areas of the animal including the head, back, belly, tail, etc. Each sample should be packed separately and labeled with the body area from which it was collected.

Fiber Evidence

Fiber evidence may be collected in the same manner as hair evidence. These methods include but not limited to picking, tape lifts, and scraping. Please refer to “Hair Evidence” section. DO NOT place fiber evidence loose in an envelope, but in a paper fold.

Fiber standards should be collected from all the sources that the victim and suspect are suspected of contacting. Submit the entire item to be used as a fiber standard. If this is not possible cut a small swatch (i.e. for a car seat), or pull random samples of fibers (i.e. for carpets). When collecting fiber standards from a vehicle, be sure to collect samples from all areas which may have transferred fibers (i.e. front and rear floorboard carpeting, all mats, front and rear seat upholstery and any trunk liners). These areas may appear the same but may be manufactured differently from each other and laboratory analysis may be needed to tell them apart.

Adhesive Tape

- Tape fragment from the crime scene and/or victim must be packaged in sealed and labeled separate parcels preferably affix on the plastic sheets.
- The reference tape fragment or tape roll recovered from the suspect in a separate sealed and labeled parcel for comparison.

Explosive Evidence (Pre-blast/Post blast)

A. General Considerations

- 2 - 5 gram of sample from the suspected explosive material/ bomb device should be received.
- In case of detonating cords like prima cord and safety fuse, only 3-6 inches of the sample cut from the prima cord must be submitted.
- Intact detonators, hand grenades, suicide jackets, mines must not enter to the lab premises and are strictly prohibited to receive.
- Only the 2 - 5 gram chemical/suspected explosive material recovered from the explosive devices [Intact detonators (0.1-0.5 gram), hand grenades, suicide jackets, mines] and sealed in a labeled parcel should be submitted for analysis.
- No trace evidence case must be sent to PFSA through courier service, all the case evidence item must be submitted by the authorized person nominated by the submitting agency along with the clear information related to the evidence samples being submitted to PFSA.
- Debris collected from the blast seat/blast scene like soil, metal parts, suspected bomb device parts like deformed metal pieces, plastic pieces, ball bearings, jagged fabric, broken time devices etc. in SEPERATE sealed and labeled parcels clearly mentioning the location from where the evidence was collected.
- Intact soil sample mentioning as “control” taken out from the uncontaminated nearby place at blast scene, in separate sealed and labeled parcel must be submitted.

B. Essential properties of Container Used for Packaging

- Unused
- Airtight (For all Fire Debris and most Explosives or Chemical Items)
- Clean–no hydrocarbon or other chemical residue
- Inert–will not break down when heated or in contact with solvents

- Will not promote a static electrical charge (For Explosives)

C. Seals

- A clean seal is essential.
- Containers must be completely sealed to prevent any passage of vapors or contaminants into or out of the container. Be certain, can lids are tight all the way around. For plastic bags, they must be heat sealed completely with no flaws in the seam if they are being used for fire debris samples.
- Tamper evident tape (tamper proof tape) must be placed across the container lid/seam in such a manner that the item cannot be partially or completely opened without tearing the tape.
- Seals and Tape must be dated with case number and exhibit number and initialed/signed by the investigator of record. The date of the seal should also be included.

Acid Examination

- The suspected container of acid recovered from the place of occurrence or from the possession of the suspect/accused, making it airtight/leak proof or pack in the air tight container.
- Suspected liquid 15-20 ml in a sealed and labeled airtight container.
- Affected clothes of victim in a sealed and labeled airtight container.
- Debris from the place of occurrence suspected to be having traces of acid spillage in sealed and labeled airtight clean containers, clearly mentioning the place from where the sample was taken out.
- Biological samples like skin, flesh, human remains, blood etc. are not accepted in Trace Chemistry Department.

Fire/Arson Cases:

- The suspected container of the ignitable liquid recovered from the place of occurrence or from the possession of the suspect making it air tight or placing in the airtight clean metal paint cane.
- Suspected ignitable liquid 15-20 ml in a sealed and labeled airtight container.
- Semi burnt cloth of victim placed in a sealed and labeled airtight metal container.
- Debris from the place of occurrence suspected to have traces of ignitable liquid residues packaged in sealed and labeled airtight clean metal containers.
- Control sample from intact place like soil debris, intact carpet piece etc. for comparison purpose.
- Packaging must be Unused.

- Airtight clean (no hydrocarbon or other chemical residue) metal paint can or heat resistant nylon bags. (For all Fire Debris and most Explosives or Chemical Items)
- Clean–Inert–will not break down when heated or in contact with solvents

Gunshot Primer Residue

Sample Required

- Pure carbon adhesive stubs dabbed from both hands of the shooter preferably one stubs from each hand including back and palms in a sealed and labeled parcel.
- No cotton swabs or hand washes from the hands of the shooter are accepted.
- At least 2 Adhesive carbon stubs dabbed from (1: right hand & Palm, 2: left hand back & palm)
- GSR sampling must be done within 4- 6 hours of shooting. After that, GSR sample may not be present for detection and identification.
- HAND WASH is not suitable for GUNSHOT PRIMER RESIDUE analysis.

Footwear/Tire Impression Comparison

- The footwear mold/cast from the place of occurrence placed in a sealed and labeled cardboard box.
- The mold of the suspect’s shoes preferably along with the shoes of the suspect in a separate sealed and labeled cardboard box.
- Tire track mold/cast from the place of occurrence in a sealed and labeled cardboard box.
- The reference mold of the tires preferably along the tires of the vehicle.
- Bare foot impressions are not accepted in Trace Chemistry Department.

Evidence Collection Instructions

Packaging Type	Case Type
Plastic bags or Ziplocs	A non-biological material such as powder/ explosive
Metal cans/ nylon bags	Arson evidence (burnt victim’s clothing)
Glass vials	Explosive/suspected powder/ignitable liquids/ acids/bases
Paper folds	Hairs, fibers, minute glass particles,paint chips
Cardboard boxes	(Physical Match cases) knives, large pieces of glass, plastic and vehicle paint from the victim’s clothing.

Pure carbon adhesive stub	Primer gunshot residue collection

Toxicology

Sample must be submitted in preservative and amount as described below.

For Medico-legal cases (MLCs)

1. **Blood:** 10 mL, preserved with sodium fluoride and potassium oxalate, mixed in the ratio 1:3. 20 mg of this mixture is sufficient for preservation of 10 ml of blood.
2. **Urine:** 20-50 mL, without any preservative.
3. **Gastric Lavage:** Minimum 20 mL, First undiluted portion without preservative.

For Postmortem Cases

1. Blood: 50-100 mL, preserved with sodium fluoride and potassium oxalate, mixed in the ratio 1:3.
   - 100-200 mg of this mixture is sufficient for preservation of 50-100 ml of blood.
2. Urine: Shall be submitted all available without preservative.
3. Stomach contents: Shall be submitted all available without preservatives.
4. Liver: Not more than 100 grams preserved in saturated saline.
5. Spleen: Not more than 100 grams preserved in saturated saline if Carbon monoxide poisoning is suspected.
6. Abdominal paste: Only in exhumation if above mentioned samples are not available (minimum 100 grams shall be submitted), without preservative.
7. Hair: Accepted only in chronic drug exposure (Hair cluster (pulled or collected as near to scalp as possible) having thickness of a pencil shall be submitted).

Collection, Preservation and Transport of Evidence

Collect toxicology samples as soon as possible after the offense, in death cases before embalming where applicable. Pack specimens in well-sealed, leak-proof containers, all samples must be collected in separate containers. For most specimens, disposable hard plastic or glass tubes are recommended. Blood tubes should be sealed and kept cold, but do not freeze. Never expose specimens to hot temperatures.

Labeling

For a valid chain of custody, all items of evidence must be labeled with the following information:

- Name of victim or suspect.
- Case number.

- Type of specimen (i.e., Blood, Urine).
- Site of collection (i.e., Femoral, Heart).
- Amount of specimen.
- Time and date of collection.
- Name(s) of the medical examiner or person collecting the sample

Finally, tamper-resistant tape with the collector's initials and the collection date should be placed over the specimen lid and container to document specimen integrity. Alternatively, all the samples collected for a given case may be placed in a tamper evident container labeled with the case number and name.

Narcotics

- Evidence must be packed in sealed, neat plastic bag, cloth wrap, paper envelop or box.
- Liquid samples should be packed in leak proof sealed bottle.
- All samples must be packed separately.
- Seal markings on evidence must be same as mentioned on FIR copy.
- Wet plant material should be placed in paper envelop or paper fold to prevent deterioration.
- FIR copy, Analysis Request Letter (e.g. DPO/CPO letter or equivalent) and Road Certificate etc. should be submitted with the evidence.
- The court's order for analysis is the least requirement for the cases submitted by courts.
- FIR # and evidence description should be same on all documents and evidence packing.
- Evidence should be submitted by the Investigation Officer or by the person whose name is written on Road Certificate as submitting person.

DNA & Serology

1 - Definition

Biological evidence refers to the samples of biological material such as hair, tissue, bones, teeth, blood, semen, or other bodily fluids. Evidence items containing one or more of the aforementioned biological material are also treated as biological evidence.

2 - Objective

The objective of effective collection, packaging and transport of biological material for forensic DNA analysis is to ensure the safety of personnel handling the evidence, preserve the integrity and quality of biological material and avoid its contamination, premature destruction or degradation.

3 - General Guideline for the Collection, Packaging and Transport of Biological Evidence

3.1 - Biological Evidence Safety and Handling

Individuals handling any evidence should assume that all of it might contain potentially hazardous biological material. It is not possible to determine if every bodily fluid or stain collected from crime scenes is contaminated with bloodborne pathogens; therefore, all bodily fluids and tissues are presumed to be contaminated. Common diseases/viruses caused by exposure to bloodborne pathogens include hepatitis and human immunodeficiency virus (HIV). These raise the most concern because of the potential for lifelong infection and the risk of death associated with infection once an individual is exposed.

1. The appropriate use of personal protective equipment (PPE) is recommended to protect the individual and the evidence from cross-contamination. PPE includes disposable gloves, disposable overalls, laboratory coats, masks, and eye protection.
2. PPE should be used in every situation in which there is a possibility of exposure to blood or infectious diseases. Gloves and protective clothing should be worn when conducting medico legal examination, autopsy or collection of biological evidence, handling soiled materials or equipment, and cleaning up spills of biologically hazardous materials. Face protectors, such as splash goggles, should be worn to protect against items that may splash, splatter, or spray.
3. PPE must be clean and in good repair. PPE that is torn or punctured, or that has lost its ability to function as an effective barrier, should not be used. Disposable PPE should not be reused under any circumstances. While using PPE, individuals should not touch their eyes or nose with gloves.
4. PPE must be removed when it becomes contaminated and before leaving the work area. Used protective clothing and equipment must be placed in designated areas for storage, decontamination, and disposal.
5. Dried blood or other dry potentially infectious material should not be assumed to be safe. PPE should be used when handling these items.
6. When wet material is spilled, the area containing blood or other potentially infectious material should be covered with paper towels or rags, covered with a disinfectant solution (10 % bleach solution), left for at least 10 minutes, and removed. Materials should then be placed in a waste disposal bag designated for biohazardous material. Appropriate PPE should be used throughout this process.

7. Biological evidence packages must be appropriately labeled to indicate that they contain biological material, which may potentially be infectious so that other individuals could avoid the risk of exposure or contamination of the evidence.
8. Any accidental direct exposure to the biological evidence must immediately be reported to an appropriate healthcare provider.

3.2 - Packaging of Biological Evidence

1. Use paper bags, manila envelopes, cardboard boxes, and similar porous materials for the packaging of all biological evidence.
2. Do not use glass bottles, plastic jars, metal containers, polythene bags or other such like non porous materials for the packaging of biological evidence. Bacterial growth or mold can irreversibly damage and degrade the biological material in such like packaging.
3. Package evidence and seal the container to protect it from loss, cross-transfer, contamination, and/or deleterious change.
4. For security purposes, seal the package in such a manner that opening it causes obvious damage or alteration to the container or its seal.
5. Evidence tape or clear scotch tape may be used to seal evidence. Mark across the seal with the sealer's identification or initials and the date. Signature of the sealer should be inscribed on the seal such that half of the signature are on the tape and the other half is on the envelope or container.
6. Package each item separately and avoid comingling items to prevent cross-contamination.
7. Each evidence item packaging must be labeled bold and clear to indicate BIOLOGICAL EVIDENCE.
8. At a minimum, mark each package with a unique identifier, the identification of the person who collected it, and the date of collection. The unique identifier should correspond to the item description noted on the PMR or MLC, police docket or doctor's request for test and the road certificate. Packaging should also be labeled to indicate the unique identity of the evidence within. For example, the name of patient, FIR Number, MLC Number or PMR Number must be clearly labeled on the packaging.
9. DNA is best preserved in an air-dried, water-free environment. Water can cause instability and breakage in strands that bind DNA, which would degrade the ability to properly test. Further, the presence of water encourages the growth of yeast, mold, and bacteria, which can also degrade DNA. Therefore all biological evidence samples must be in dry form prior to packaging, temporary storage or transport. Drying wet items of evidence, such as swabs or a blood-soaked or mud-stained garment, should be the first task of anyone handling wet biological evidence once it has been collected.
10. Blood-draw samples or tissue samples may however be packaged and transported without drying. If drying wet evidence is not possible, place the evidence in an impermeable, nonporous container and place the container in an ice box or refrigerator that maintains a temperature of 2 °C – 8 °C and that is

located away from direct sunlight until the evidence can be submitted to the laboratory.

11. Unload, make safe, and place all firearms submitted into evidence for biological testing into a new cardboard gun box. As the submitting individual, seal the box and indicate on the exterior of the box that the weapon was unloaded, made safe, and may contain biological material.
12. Sharp weapons such as knives, daggers, showels etc. should be packaged in appropriate sized cardboard box or carton. As the submitting individual, seal the box and indicate on the exterior of the box that it contains SHARP weapon and may contain biological material.
13. Maintain the integrity of the item through the package documentation, including all markings, seals, tags, and labels used by all of the involved agencies. Preserve and document all packaging and labels received by or returned to the agency, because this information is critical.

4 - Guideline for the Collection, Packaging and Transport of Evidence in Sexual Assault Cases

1. PFSA Sexual Assault Evidence Collection Kits (SAECK) are designed for the effective collection of evidence from the victims of sexual assault. PFSA SAECKs must always be used for the collection and packaging of evidence in rape cases. The PFSA SAECK contains sterile swab sticks, distilled water and comb for the collection of evidence; a pair of sterile gloves to be used while collecting and handling biological material for the safety of evidence collector and prevention of samples from contamination; paper envelopes of various sizes for the packaging of evidence and tamper proof tape for appropriate sealing of evidence for submission.
2. Victim's underwear and garments worn at the time of assault should be collected and packaged in the envelope provided in the sexual assault evidence kit.
3. Evidence should be collected after a thorough evaluation of the assault and background history are obtained, if possible. Documentation typically referred to as medico legal examination report or certificate (MLC) should contain specific information about the assault, what items were collected during the exam, and personal information from the victim.
4. Prior to collection of evidentiary items, medico legal examiners must consider several factors to assist in guiding their collection and treatment efforts. These factors may include the assault activity, time elapsed since the assault, post assault activities, the age and gender of the assault victim, and mental capacity, to name a few. History written in the MLC should incorporate descriptive notes on the aforementioned factors.
5. Evidence collection should be guided by the background history, focusing specifically on the suspect's actions during the assault. However if the victim is unable to recollect a complete background history due to trauma or pre-existing mental in capacity, a full range of samples should be collected assisted by the physical assessment.

6. Additional considerations prior to sample collection must include the activities of the victim following the assault. Activities that may impact evidence collection include bathing, brushing of teeth, mouthwash, vomiting, douching, urination and defecation. Careful consideration of the assault activities and post assault activities prior to sample collection is vital. For example, the analysis of swabs collected by swabbing from areas that are kissed, licked, sucked or bit may be impacted if the victim has showered or bathed between the assault and the time of collection. The victim should therefore be carefully interviewed to record appropriate observations about these post assault activities and findings should be documented on the MLC.
7. Internal swabs such as from the vagina, mouth or rectum may still be viable for collection even after showering or bathing by the victim, dependent upon the length and thoroughness of the cleansing and time since the assault. Internal and external swabs should still be collected even if the victim has bathed, as the bathing may not have been vigorous enough to remove the fluids or DNA from the victim.
8. Potential biological evidence deposited onto a substrate such as clothing, towels, paper towels or tissue papers do not have the same time restrictions as biological evidence deposited on, or within, the victim's body. Therefore, garments and wipes etc. must always be collected and submitted for DNA analysis regardless of the time elapsed since sexual assault.
9. Evidential items are collected with the perspective of recovering as much DNA foreign to the victim as possible during the collection process. Measures should therefore be taken to concentrate the foreign material by using the fewest number of swabs necessary for the collection site.
10. If multiple swabs are used during the collection, they should be collected concurrently.
11. If swabs are not taken concurrently, then the order of the swabs collected must be noted, appropriately labeled on the swab packaging and documented. When more than one swab is collected from an area then these swabs should also be collected in a consistent fashion. For example if one moistened swab was used for evidence collection, then the second swab should also be moistened.
12. Only sterile standard cotton tip swabs, provided in PFSA SAECK or otherwise commercially available, can be used for collection of evidence from the body of victim. Homemade swabs and cotton balls etc. should never be used for evidence collection.
13. Swabs should always be properly air dried prior to packaging. Swabs should never be packed in any liquid or preservative.
14. Garments, if blood stained or wet, should be dried prior to packaging.
15. Semen has a limited post-coital DNA persistence time to reside on the surface of the body and within a body tract. The maximum recommended time frames for evidence collection in the cases of sexual assault are as under:

Type of Assault	**Maximum Post Coital Time Duration for Evidence Collection**
Vaginal	Up to 120 hours (5 days)
Anal	Up to 72 hours (3 days)

Oral	Up to 24 hours (1 day)
Bite marks	Up to 96 hours (4 days)

16. Alleged assaults that may have resulted in deposition of semen externally (victim's clothing, bedding, etc.) should also result in evidence collection because semen will remain indefinitely on these items as long as they are unwashed.

5 - Guideline for the Collection, Packaging and Transport of Biological Evidence in Dead Body Identification Cases

Sample Submission Form for Dead Body Identification through DNA Profiling

6 - Reference Samples

Buccal swab are collected at PFSA Lahore as standard reference for DNA profiling, in accordance with the ERU SOPs.